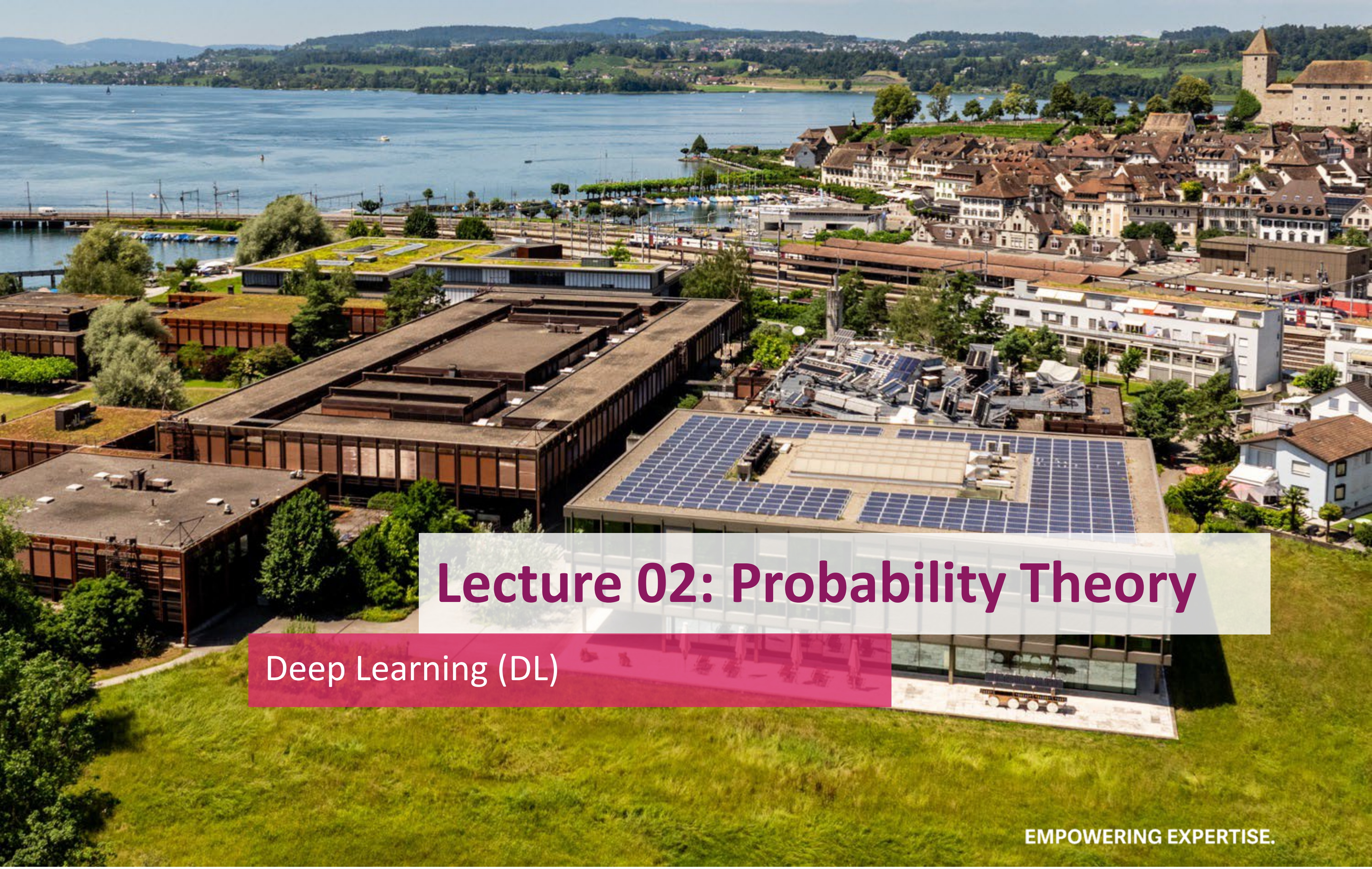

Lecture 02: Probability Theory

Deep Learning (DL)

ICAI

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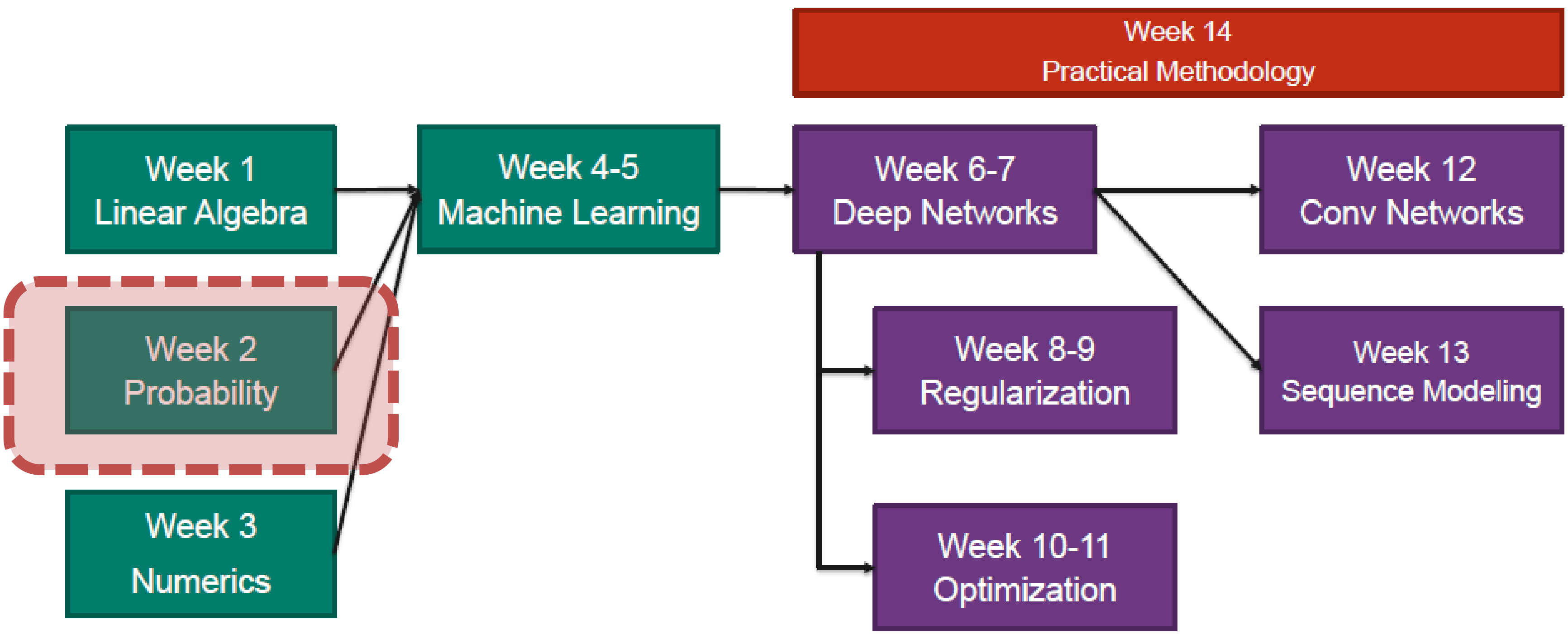
Overview of DL-Lecture



Overview of this lecture

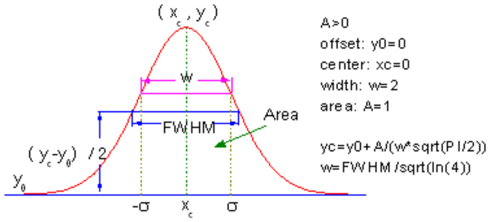
- Basics of probability
- Probability distributions
- Important functions
- Information theory

Random Variables

- A **random variable** (rv) is a variable that can take on **different values** randomly
  - This does not mean, that every value is equally likely!
- There are two types of rv

Type of rv	Values of rv	Description of rv with …
Continuous	Real numbers	Probability density function (pdf)
Discrete	Discrete set	Probability mass function (pmf)

- Example: **Random Noise** is a continuous rv
  - Often (but not always) noise follows a normal distribution



Discrete random variables

- **Discrete rv's** have a **probability mass function** (PMF)
  - Notation: $P(x)$ or $P_x(x)$
  - P(x) and P(y) are NOT the same
  - A PMF is directly the probability of x

$$\mathrm{x} \sim P(\mathrm{x})$$

- **Properties** of a PMF

1. Domain of $P(x)$: all possible states of x
2. $0 \leq P(x) \leq 1$
3. $\sum P(x) = 1$

- Note:
  - $P(x) = 0$ does not mean impossible – only highly improbable
  - $P(x) = 1$ does not mean guaranteed – only highly probable

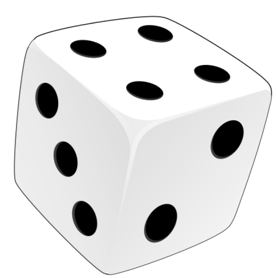
Discrete random variables

- Example: **discrete uniform distribution**
  - Random variable *x* with *k* different states x_i
  - Each state is **equally likely**
  - Resulting PMF:

$$\boxed{P(\mathrm{x} = x_i) = \frac{1}{k}}$$!

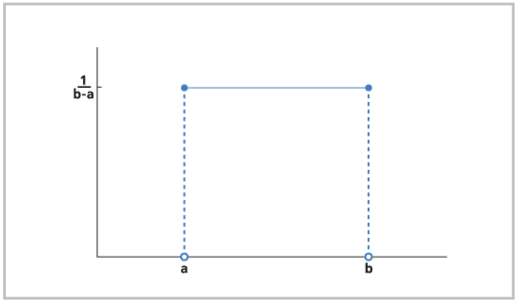
- **Properties**
  1. Domain = all possible states
  2. $0 \leq P(x) \leq 1$
  3. $\sum P(x) = k\,\frac{1}{k} = 1$

- Examples
  - Coin → k=2
  - Dice → k=6



Continuous random variables

- **Difficulty**: can take on an **infinite number of values**
  - ➔ a PMF would always be 0
- Instead of a PMF, a **probability density function** (**PDF**) is used
  - PFM: probability
  - PDF: **probability per area = likelihood**
  - **Notation:** lowercase *p(x)* or $p_x(x)$



$$x \sim p(x)$$

- **Properties**

1. Domain of *p(x)*: all possible values of *x*
2. $p(x) \geq 0$ (but not ≤ 1)
3. $\int p(x)\,dx = 1$

Continuous random variables

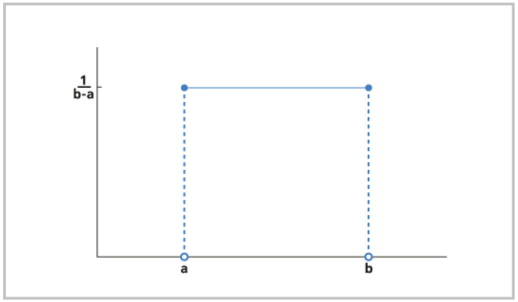
- **Probabilities** only make sense **over an interval**
  - Because likelihood = probability / area, a probability is obtained by integrating

$$P(a \leq x \leq b) = \int_{[a,b]} p(x)dx$$!

- Example: **continuous uniform distribution**
  - Each value in the interval *[a, b]* is equally likely
  - Notation

$$\mathrm{x} \sim U(a,b)$$

$$u(x;a,b) = \frac{1}{b-a}$$



Joint and Marginal Probability

- Joint PDF or PMF
  - 2-D (or multi-dimensional) PDF or PMF over multiple variables

$$P(\mathrm{x}, \mathrm{y})$$

$$p(x, y)$$

- Marginal probability
  - "**Marginalization**" = obtain PMF or PDF of a single random variable from a joint PMF or PDF
  - Marginal probability = same thing as PMF or PDF of that variable

$$P(\mathrm{x} = x) = \sum_y P(\mathrm{x} = x, \mathrm{y} = y)$$

$$p(x) = \int p(x, y)dy$$

Conditional Probability

- **Goal**: describe probability of one random variable, given one knows another random variable

$$P(y \mid x)$$

- **Examples**
  - P(sales | advertisement budget)
  - P(cat | pixels of an image)

- Definition

$$P(\mathrm{y} = y \mid \mathrm{x} = x) = \frac{P(\mathrm{y} = y, \mathrm{x} = x)}{P(\mathrm{x} = x)}$$

Chain Rule of Conditional Probabilities

- The joint probability over multiple variables can decomposed into a chain of conditional probabilities
  - This is obtained by applying the definition shown on the right multiple times

$$\boxed{P(\mathrm{x}^{(1)}, \ldots, \mathrm{x}^{(n)}) = P(\mathrm{x}^{(1)}) \Pi_{i=2}^{n} P(\mathrm{x}^{(i)} \mid \mathrm{x}^{(1)}, \ldots, \mathrm{x}^{(i-1)})}\ \textbf{!}$$

$$P(\mathrm{y} = y \mid \mathrm{x} = x) = \frac{P(\mathrm{y} = y, \mathrm{x} = x)}{P(\mathrm{x} = x)}$$

- Easier example: 3 rv's
  - Step 1: treat b,c together:

$$P(\mathrm{a}, \mathrm{b}, \mathrm{c}) = P(\mathrm{a} \mid \mathrm{b}, \mathrm{c}) \boxed{P(\mathrm{b}, \mathrm{c})}$$

  - Step 2: calculate P(b,c)

$$\boxed{P(\mathrm{b}, \mathrm{c})} = P(\mathrm{b} \mid \mathrm{c}) P(\mathrm{c})$$

  - Combine everything!

$$P(\mathrm{a}, \mathrm{b}, \mathrm{c}) = P(\mathrm{a} \mid \mathrm{b}, \mathrm{c}) P(\mathrm{b} \mid \mathrm{c}) P(\mathrm{c})$$

Bayes' Rule

- Goal: calculate *P(x|y)* from *P(y|x)*
  - Very common problem

$$\boxed{P(\mathrm{x} \mid \mathrm{y}) = \frac{P(\mathrm{x})P(\mathrm{y} \mid \mathrm{x})}{P(\mathrm{y})}}$$!

- Example: Medical test
  - We are interested in P(sick | test)
  - P(test | sick) is easy to obtain
  - P(sick) can also easily be obtained

$$P(sick \mid test) = \frac{P(test \mid sick)\, P(sick)}{P(test)}$$

- ***P(y)*** is often unknown, but can easily be calculated with the rule of **total probability**

$$\boxed{P(\mathrm{y}) = \sum_x P(\mathrm{y} \mid x)P(x)}$$!

Statistical independence

- Two random variables x and y are called **statistically independent**, if:
  - Joint probability = product of marginal probabilities

$$\boxed{p(\mathrm{x}=x,\mathrm{y}=y)=p(\mathrm{x}=x)p(\mathrm{y}=y)}$$!

- This results in the **conditional probabilities**
  - Hence, knowing **x helps nothing for describing y** – and vice versa

$$p(x|y)=\frac{p(x,y)}{p(y)}=\frac{p(x)p(y)}{p(y)}=p(x)$$

$$p(y|x)=\frac{p(x,y)}{p(x)}=\frac{p(x)p(y)}{p(x)}=p(y)$$

- Short-hand notation

$$\boxed{\mathrm{x}\perp\mathrm{y}}$$

Conditional Independence

- Two random variables x and y are **conditionally independent** given a third random variable z, if
  - If z is known, then x and y are independent.
  - If z is unknown, they are not independent!

$$\boxed{p(\mathrm{x}=x,\mathrm{y}=y \mid \mathrm{z}=z)=p(\mathrm{x}=x \mid \mathrm{z}=z)p(\mathrm{y}=y \mid \mathrm{z}=z)}$$

!

- Short-hand notation

$$\boxed{\mathrm{x} \perp \mathrm{y} \mid \mathrm{z}}$$

- **Application** in ML: **Naïve Bayes**
  - Features (inputs x) are considered statistically independent, given the true class is known

Expectation

- Expectation = **weighted average** over P(x)

- Discrete case: **sum**

$$\boxed{\mathbb{E}_{\mathrm{x}\sim P}[f(x)] = \sum_x P(x) f(x)}\ !$$

- Continuous case: **integral**

$$\boxed{\mathbb{E}_{\mathrm{x}\sim p}[f(x)] = \int p(x) f(x) dx}\ !$$

- The full notation is not used often, instead the following short-hand notations are used
  - Obviously, x follows some pdf p
  - If there is only one random variable, clearly the expectation is over that.

$$\mathbb{E}_{\mathrm{x}}[f(x)]$$

$$\mathbb{E}[f(x)]$$

Variance

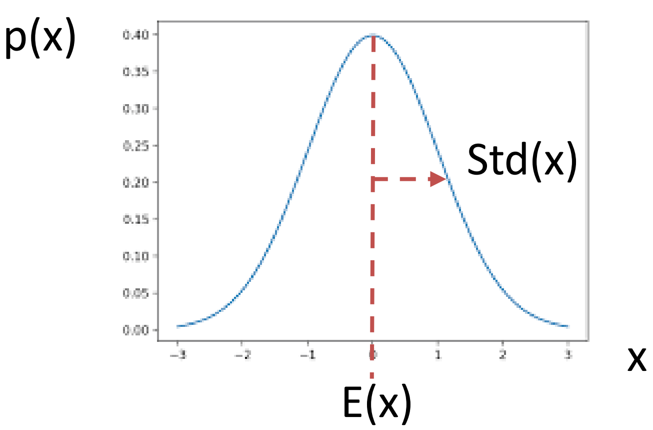
- The **variance** σ^2 measures the variability around the expectation
  - Low variance ➔ all values x will be close around the expectation E(x)
  - High variance ➔ values x can vary much from the expectation E(x)

$$\mathrm{Var}(f(x)) = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])^2\right]$$!

- The **standard deviation** σ is the square root of the variance
  - Advantage: same unit as f(x)

$$\mathrm{Std}\big(f(x)\big) = \sqrt{\mathrm{Var}(f(x))}$$!

- Example: Gaussian PDF



Covariance and Correlation

- Covariance σ_{xy} Measure of **linear relation** between **two random variables** *x* and *y*
  - **Sign**: positive or negative relationship
  - **Absolute value**: scale of relationship and of individual variability
  - Note: outer expectation is over both x and y, hence over their joint distribution $p(x, y)$

!

$$\boxed{\mathrm{Cov}(f(x), g(y)) = \mathbb{E}\left[(f(x) - \mathbb{E}\left[f(x)\right])\left(g(y) - \mathbb{E}\left[g(y)\right]\right)\right]}$$

$$E_{x,y\sim p(x,y)}[\ldots]$$

- **Correlation**: covariance divided by the standard deviations
  - Range: [-1, +1]
  - ➔ independent of individual variabilities of x and y

$$\boxed{\mathrm{Corr}\big(f(x), g(y)\big) = \frac{\mathrm{Cov}(f(x), g(y))}{\mathrm{Std}\big(f(x)\big)\mathrm{Std}(g(Y))}}$$

!

Covariance matrix

- **Extension** of variance and covariance **to random vectors**
- **Definition** of covariance matrix

$$\boxed{\mathrm{Cov}(\mathbf{x})_{i,j} = \mathrm{Cov}(\mathrm{x}_i, \mathrm{x}_j)} \ \textbf{!}$$

- **Example**: covariance matrix of a 3-d random vector

$$\mathrm{Cov}(\boldsymbol{x}) = \begin{bmatrix} \mathrm{Cov}(x_1, x_1) & \mathrm{Cov}(x_1, x_2) & \mathrm{Cov}(x_1, x_3) \\ \mathrm{Cov}(x_2, x_1) & \mathrm{Cov}(x_2, x_2) & \mathrm{Cov}(x_2, x_3) \\ \mathrm{Cov}(x_3, x_1) & \mathrm{Cov}(x_3, x_2) & \mathrm{Cov}(x_3, x_3) \end{bmatrix}$$

- The **covariance matrix** is **symmetric** because

$$\mathrm{Cov}(x_i, x_j) = \mathrm{Cov}(x_j, x_i)$$

- The diagonal contains the variances

$$\mathrm{Cov}(x_i, x_i) = \mathrm{Var}(x_i)$$

Overview of this lecture

- Basics of probability
- Probability distributions
- Important functions
- Information theory

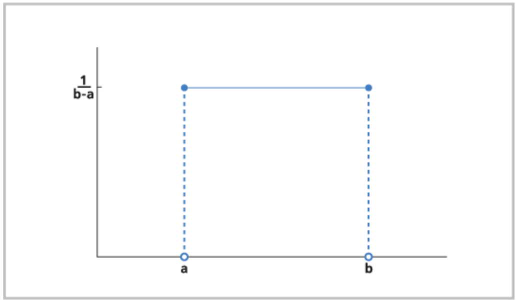
Uniform Distribution

- Repetition from before
  - In general: **all possible values** are **equally likely**
- Discrete uniform distribution
  - *k* possible **states** are **equally likely**

$$P(\mathrm{x} = x_i) = \frac{1}{k}$$

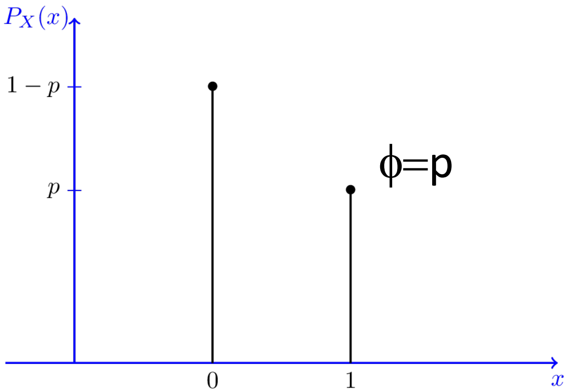
- Continuous uniform distribution
  - All **values in interval [a, b]** are possible and **equally likely**

$$p(x) = \frac{1}{b - a}$$



Bernoulli Distribution

- Discrete random variable with **two states**: 0 and 1
  - ➔ for **binary** random variables



- Single parameter ϕ or p, which is defined as:

$$\boxed{P(\mathrm{x}=1)=\phi} \; !$$

  - Because only two states, $P(x=0)$ can be calculated as

$$\boxed{P(\mathrm{x}=0)=1-\phi} \; !$$

  - Both can be elegantly combined into one equation

$$P(\mathrm{x}=x)=\phi^{x}(1-\phi)^{1-x}$$

- Properties of the Bernoulli distribution

$$\mathbb{E}_{\mathrm{x}}[\mathrm{x}]=\phi$$

$$\mathrm{Var}_{\mathrm{x}}(\mathrm{x})=\phi(1-\phi)$$

Multinoulli Distribution

- Extension of Bernoulli to **discrete variables** with ***k* different states**
- Parameter vector p with *k-1* probabilities where

$$\boldsymbol{p} \in [0,1]^{k-1}$$

$$p_i = P(x = x_i)$$

- **Note**: for *k* states, only *k-1* parameters are required, as they must sum to 1
  - The final probability can be calculated with
  - For Bernoulli with 2 states, only 1 parameter was required

$$p_k = 1 - \sum_{i=0}^{k-1} p_i = 1 - \mathbf{1}^T \boldsymbol{p}$$

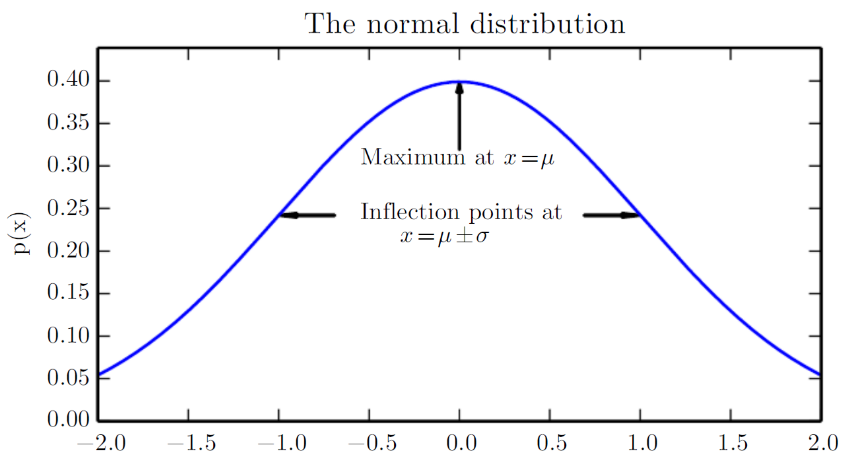
- **Application**: describe probabilities of multiple classes

Gaussian Distribution

- Very common variable for uniform distributions: **Gaussian**
  - Also called "**normal" distribution**, hence the notation $\mathcal{N}$

$$\boxed{\mathcal{N}(x;\mu,\sigma^2) = \sqrt{\frac{1}{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)}$$!

- Two parameters
  - μ the **mean** (expected value) and
  - σ the **standard deviation** (σ^2 the variance)



DL, Fig. 3.1

Gaussian Distribution

- **Alternative parameterization**
  - β is called the **precision**
  - Note: β is simply $1/\sigma^2$ ➔ still the same distribution
  - Advantage: no inverse required ➔ more computationally efficient, especially in multi-dimensional cases

$$\mathcal{N}(x;\mu,\sigma^2) = \sqrt{\frac{1}{2\pi\boxed{\sigma^2}}}\exp\left(-\frac{1}{2\boxed{\sigma^2}}(x-\mu)^2\right)$$

$$\mathcal{N}(x;\mu,\beta^{-1}) = \sqrt{\frac{\boxed{\beta}}{2\pi}}\exp\left(-\frac{1}{2}\boxed{\beta}(x-\mu)^2\right)$$

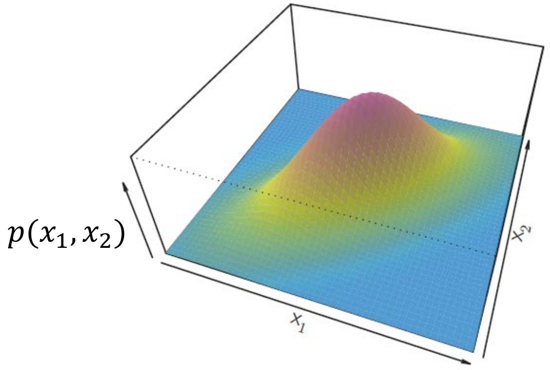
- Normal distributions are a popular because of
  - 1) The **central limit theorem**, which states that the **sum of many independent rvs** approaches a **Gaussian** distribution
  - 2) For a **given variance**, the Gaussian distribution has the **largest uncertainty** (differential entropy)

Gaussian Distribution

- Multidimensional Gaussian
  - These concepts can be extended to **n-dimensional random vectors**

$$\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sqrt{\frac{1}{(2\pi)^n \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$!

- Parameters
  - Mean vector $\boldsymbol{\mu}$
  - Covariance matrix $\boldsymbol{\Sigma}$



ISLP, Fig. 4.5

Gaussian Distribution

- Alternative parametrization
  - The **precision matrix β** can be used instead of the covariance matrix Σ
  - Advantage: **no matrix inverse** required

$$\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu},\boldsymbol{\Sigma}) = \sqrt{\frac{1}{(2\pi)^n \boxed{\det(\boldsymbol{\Sigma})}}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boxed{\boldsymbol{\Sigma}^{-1}} (\boldsymbol{x}-\boldsymbol{\mu})\right)$$

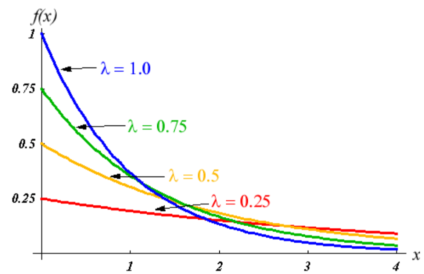
$$\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu},\boldsymbol{\beta}^{-1}) = \sqrt{\frac{\boxed{\det(\boldsymbol{\beta})}}{(2\pi)^n}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boxed{\boldsymbol{\beta}} (\boldsymbol{x}-\boldsymbol{\mu})\right)$$

- For **high-dimensional vectors**, the **covariance matrix** can become very **large**
  - Sometimes, the covariance matrix is **constrained** to be **diagonal**
  - **Assumption**: dimensions of x uncorrelated

- Isotropic Gaussian
  - “isotropic” = same in all directions
  - Same variance for all dimensions, no covariance

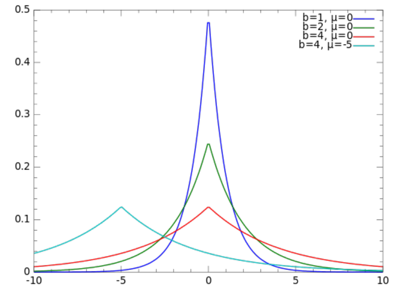
Exponential and Laplace Distribution

- **Idea**: 0 is more likely than non-zero values
- **Exponential distribution**
  - Function $\mathbf{1}_{x\geq 0}()$ is a step function: 0 for negative, 1 for positive inputs
  - Parameter λ controls steepness and hence height of peak at 0
- **Laplace distribution**
  - Related to Gaussian, but with absolute value instead of square
  - Peak at mean μ
  - Parameter γ controls steepness (and variance)



$$p(x;\lambda) = \lambda \mathbf{1}_{x\geq 0} \exp\left(-\lambda x\right)$$!

$$\text{Laplace}(x;\mu,\gamma) = \frac{1}{2\gamma} \exp\left(-\frac{|x-\mu|}{\gamma}\right)$$!



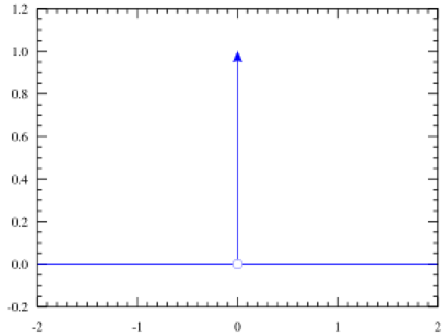
Dirac Distribution

- **Dirac function** / dirac impulse
  - Infinitely high value in an infinitely small range
  - Interesting fact: area under dirac function (integral) = 1, hence

$$\delta(x) = \begin{cases} \infty, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

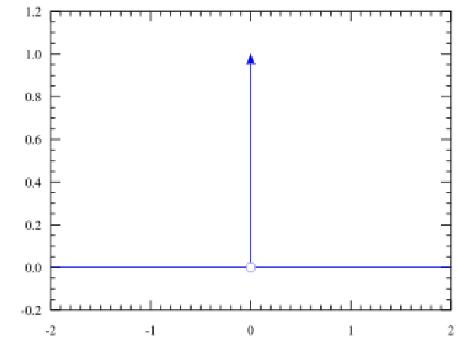
$$\int_{-\infty}^{\infty} \delta(x)dx = 1$$

- The function is 0 everywhere, except at 0, there it goes to infinity
    - But integrating over 0 results in 1
  - Hence, this allows the replacement of pmf by pdf, simply put a Dirac delta function at all the locations where the pmf has probability
  - For example, the way this pdf p(x) is defined, the probability mass of 1 is all at μ, hence the pdf is zero everywhere, except at μ, where it is infinite. But the probability at μ is 1, since it is the integral over an infinitesimal interval around 0



$$\boxed{p(x) = \delta(x - \mu)}$$!

Dirac Distribution



- **Dirac function** / Dirac impulse
  - Infinitely high value in an infinitely small range
  - Interesting fact: area under Dirac function (integral) = 1, hence

$$\delta(x) = \begin{cases} \infty, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

- The **Dirac distribution** is given by
  - Only one value μ is possible and has a likelihood of ∞

$$\boxed{p(x) = \delta(x - \mu)} \; !$$

- **Multiple Dirac distributions** can be combined, for example
  - Three possible values: -1, 0, +1, with probabilities of 0.25, 0.5, 0.25
  - ➔ like a PMF over a discrete set of values, but in continuous form

$$p(x) = 0.25\delta(x+1) + 0.5\delta(x) + 0.25\delta(x-1)$$

Empirical Distribution

- **Goal**: describe probability distribution of measured data

- Given a set of ***m*** **measured values**

$$\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$$

  - These were really **measured** ➔ **high probability** assigned to all
  - All **other values** were not measured ➔ **low (0) probability** assigned to all

- **Empirical distribution**
  - **Dirac** at each measured value
  - ➔ the **true distribution** that was **measured**

$$\boxed{\hat{p}(\boldsymbol{x}) = \frac{1}{m} \sum_{i=1}^{m} \delta(\boldsymbol{x} - \boldsymbol{x}^{(i)})}$$

!

Overview of this lecture

- Basics of probability
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- Important functions
- Information theory

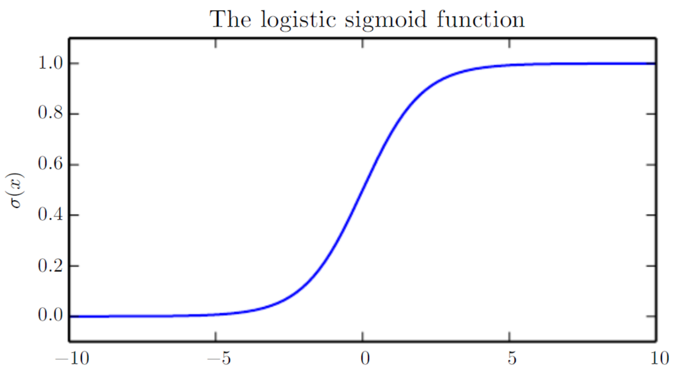
Logistic Sigmoid Function

- **Definition** of logistic sigmoid
  - Note: $\boldsymbol{\sigma}(\boldsymbol{x})$ is the **sigmoid function**, but $\boldsymbol{\sigma}$ is the **standard deviation** – these are entirely different!

$$\boxed{\sigma(x) = \frac{1}{1+\exp(-x)}} \; !$$

$$= \frac{\exp(x)}{1+\exp(x)}$$

- **Applications**
  - Logistic regression (ML)
  - Estimate parameter ϕ of a **Bernoulli distribution**
  - Estimate any output that must be between 0 and 1



DL, Fig. 3.3

- Note: **sigmoid saturates** for very high and very low inputs
  - Desirable ➔ bounds output to [0,1]
  - But: will be problematic later on

Softplus Function

- Plus function x^+
  - Cuts away the negative part
  - Also called the "**Rectified Linear**" function

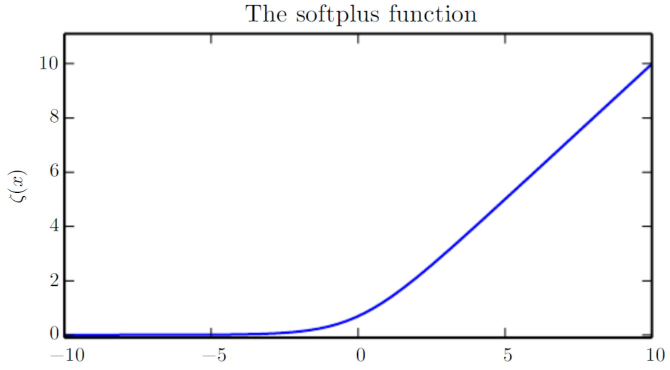
$$x^+ = \max(0, x)$$

- Softplus function $\zeta(x)$
  - "Soft" version of the plus function
  - No sharp edge at 0, but a **smooth transition**
  - Range of output: $(0, \infty)$

$$\boxed{\zeta(x) = \log\left(1 + \exp(x)\right)}$$

!

- Applications
  - Estimating the parameters β or σ of a **Gaussian distribution**
  - Estimate any output that must be **positive**



DL, Fig. 3.4

Properties of Sigmoid and Softplus

- Useful algebraic relationships
  - The sigmoid can also be written as shown on the right (exp(0)=1)
  - Furthermore, its derivative can be expressed by the function itself
    - Not that surprising, since the sigmoid consists of exponential functions which are their own derivatives
  - The sigmoid has an interesting symmetry and the log of the sigmoid results in the flipped negative softplus
  - The derivative of the softplus turns out to be the sigmoid, i.e., the integral of the sigmoid results in the softplus

$$1 - \sigma(x) = \sigma(-x)$$

$$\zeta(x) - \zeta(-x) = x$$

$$\log \sigma(x) = -\zeta(-x)$$

$$\frac{d}{dx}\zeta(x) = \sigma(x)$$

$$\zeta(x) = \int_{-\infty}^{x} \sigma(y)dy$$

Properties of Sigmoid and Softplus

- Useful properties of the sigmoid and softplus

$$1 - \sigma(x) = \sigma(-x)$$

$$\zeta(x) - \zeta(-x) = x$$

- The sigmoid and softplus are closely related

$$\log \sigma(x) = -\zeta(-x)$$

$$\frac{d}{dx}\zeta(x) = \sigma(x)$$

$$\zeta(x) = \int_{-\infty}^{x} \sigma(y) dy$$

Applying functions to random variables

- Given a random vector ***x*** and a **deterministic function** *g(**x**)*, then ***y*** will also be a random vector

$$\boxed{\boldsymbol{y} = g(\boldsymbol{x})} \;!$$

- If the distribution $p_x(\boldsymbol{x})$ is known, **how can we calculate $\boldsymbol{p_y(y)}$?**

- Intuitive (but **wrong**) approach:

$$\boldsymbol{x} = g^{-1}(\boldsymbol{y})$$

$$p_x(\boldsymbol{x}) = p_x\left(g^{-1}(\boldsymbol{y})\right) \neq p_y(\boldsymbol{y})$$

  - Example:

$$x \sim U(0,1)$$

$$y = f(x) = \frac{x}{2}$$

  - The wrong approach would lead to:

$$p_y(y) = p_x(2y)$$

  - Integrating the pdf results in:

$$\int p_y(y)dy = \frac{1}{2}$$

  - Reason: rescaling required!

Applying functions to random variables

- **Rescaling** the probability density functions **correctly**
  - **Example**: trivial, multiply $p_y(y)$ by 2
  - In general: **equation on the right** must hold for any **infinitesimally small region** dx and dy
  - ➔ **integral** over small regions of p_x and p_y **must be equal**

$$|p_y(g(x))dy| = |p_x(x)dx|$$

- **Solving** for $p_x(x)$ and $p_y(y)$ results in:

$$p_y(y) = p_x\left(g^{-1}(y)\right)\left|\frac{\partial x}{\partial y}\right|$$

$$p_x(x) = p_y(g(x))\left|\frac{\partial g(x)}{\partial x}\right|$$

- Example $x = 2y$

$$\left|\frac{dx}{dy}\right| = 2$$

  - ➔ multiply $p_y(y)$ by 2

Applying functions to random variables

- **Generalization** to random vectors

$$p_x(\boldsymbol{x}) = p_y(g(\boldsymbol{x}))\left|\det\left(\frac{\partial g(\boldsymbol{x})}{\partial \boldsymbol{x}}\right)\right|$$

- Because $g(\boldsymbol{x}) = \boldsymbol{y}$, the term $\frac{\partial g(\boldsymbol{x})}{\partial \boldsymbol{x}}$ is a **matrix**
  - This is called the **Jacobian matrix J**, which contains the **partial derivatives** of **all outputs** with respect to **all inputs**
  - **Determinant**: **measure of volume ➔** measures, how much "space" is stretched or shrinked

$$\frac{\partial g(\boldsymbol{x})}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial g_1(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial g_1(\boldsymbol{x})}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial g_m(\boldsymbol{x})}{\partial x_n} \end{bmatrix}$$

- For large vectors x and y, the Jacobian matrix can get huge!
  - Example: for *x=1000, y=1000*, the Jacobian contains *1 Mio* elements

Overview of this lecture

- Basics of probability
- Probability distributions
- Important functions
- Information theory

Information Theory

- **Information theory** is a mathematical framework that **uses probability theory** to **describe information**

- Any **measure of information** should satisfy the **three rules** below

1. **Likely events** should have **low information** content.
2. **Less likely events** should have a **higher information** content.
3. **Independent events** should have **additive** information.

Self-Information

- Given an **event x** that occurs with **probability $P(x)$**

- The **self-information $I(x)$** is defined as:
  - That is, the **information content** of (one) **event x** happening
  - Equivalently, this can be written as:

$$I(x) = \log\left(\frac{1}{P(x)}\right)$$

$$\boxed{I(x) = -\log P(x)}$$!

- **Base** of the logarithm
  - Not important, corresponds to a change in unit
  - Base 2 ➔ Unit is called "bits"
  - Base e ➔ Unit is called "nats"

Shannon Entropy

- Self-information ➔ single event
- **Shannon entropy $\boldsymbol{H(x)}$ ➔ average self-information** over an entire **distribution $\boldsymbol{P(x)}$**
  - **Expected information content** when repeatedly drawing events from $P(x)$
  - ➔ depends only on the distribution $P(x)$, hence often the notation $H(P)$ is used
- **Self-information** is the **number of bits** required for encoding one **single event**
- **Shannon entropy** is the **number of bits** required for encoding a **stream of events**

$$\boxed{H(\mathrm{x}) = \mathbb{E}_{\mathrm{x}\sim P}[I(x)] = -\mathbb{E}_{\mathrm{x}\sim P}[\log P(x)]} \; !$$

$$H(P) = H(x)$$

Shannon Entropy

- Example
  - Given a **binary random variable** (**Bernoulli** distribution) with the parameter

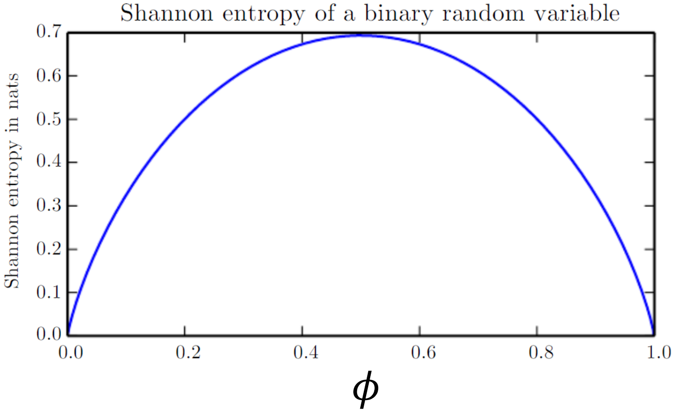
$$\phi = P(x = 1)$$

  - **If $\boldsymbol{\phi = 0}$,** x is always 0 ➔ no information content
  - **If $\boldsymbol{\phi = 1}$**, x is always 1 ➔ no information content
  - **If $\boldsymbol{\phi = 0.5}$**, then

$$I(x = 0) = \log\left(\frac{1}{0.5}\right) = \log(2)$$
$$I(x = 1) = \log\left(\frac{1}{0.5}\right) = \log(2)$$

$$H(x) = E[I(x)] = \log(2)$$

  - ➔ requires 1 bit or 0.7 nats



DL, Fig. 3.5

Kullback-Leibler (KL-) Divergence

- Measure **how different** two distributions ***P*** and ***Q*** are.

$$D_{\mathrm{KL}}(P\|Q) = \mathbb{E}_{\mathrm{x}\sim P}\left[\log \frac{P(x)}{Q(x)}\right]$$
$$= \mathbb{E}_{\mathrm{x}\sim P}\left[\log P(x) - \log Q(x)\right]$$

!

  – If $P(x) = Q(x)$, then: $D_{KL}(P \parallel Q) = E_{x\sim P}[\log(1)] = 0$

  – The higher $D_{KL}(P \parallel Q)$, the more different P and Q are

- Note: the KL-divergence is **not symmetric**

$$D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$$

  – Hence, it is **not a** mathematically valid **distance metric**

KL-Divergence and Cross-Entropy

- The **KL-divergence** can be decomposed into two parts:
  - The first term is the $\boxed{\textbf{Shannon entropy } H(P)}$
  - The second term is called the $\boxed{\textbf{cross-entropy } H(P,Q)}$ between the distributions *P* and *Q*

$$D_{\mathrm{KL}}(P\|Q) = \boxed{\mathbb{E}_{\mathrm{x}\sim P}[\log P(x)]} - \boxed{\mathbb{E}_{\mathrm{x}\sim P} \log Q(x)}$$

$$D_{KL}(P \parallel Q) = \boxed{-H(P)} + \boxed{H(P,Q)}$$

- The **cross-entropy** is defined as:

$$\boxed{H(P,Q) = -\mathbb{E}_{\mathrm{x}\sim P} \log Q(x)} \quad !$$

Information Theory

- **Meaning** of measures in information theory

Measure	Meaning
Shannon entropy $H(P)$	**Number of bits** required to send **symbols from *P*** using an **encoding optimally designed for *P***
Cross-entropy $H(P, Q)$	**Number of bits** required to send **symbols from *P*** using an **encoding optimally designed for *Q***
KL-divergence $D_{\mathrm{KL}}(P\|Q)$	***Extra* number of bits** required, when sending **symbols from P** with an **encoding optimally designed for *Q*** instead of ***P***

- Typically, in information theory, we assume:
$0 \log(0) = 0$

Information Theory

How is this useful in Deep Learning?

- Given a set of m **training data** points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$
- The **empirical data distribution** is then: $\hat{p}(\boldsymbol{x}) = \frac{1}{m} \sum_{i=1}^{m} \delta(\boldsymbol{x} - \boldsymbol{x}^{(i)})$
- The **neural network predicts** a distribution $p_{\text{model}}(\boldsymbol{x})$
- **Training process**: make $p_{\text{model}}(\boldsymbol{x})$ as similar as possible to $\hat{p}(x)$
- This can be done by **minimizing the KL-divergence** $D_{\text{KL}}(P \| Q)$
  - Or equivalently: **minimize the cross-entropy** $H(P, Q)$

Overview of this lecture

- Basics of probability
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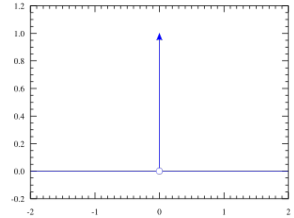
Summary: Probability Theory

Type of rv	Values of rv	Description of rv with …
Continuous	Real numbers	Probability density function (pdf), *p(x)*
Discrete	Discrete set	Probability mass function (pmf), *P(x)*

- For continuous rv $$P(a \le x \le b) = \int_{[a,b]} p(x)dx$$
- Bayes rule $$P(\mathrm{x} \mid \mathrm{y}) = \frac{P(\mathrm{x})P(\mathrm{y} \mid \mathrm{x})}{P(\mathrm{y})}$$ where $$P(\mathrm{y}) = \sum_x P(\mathrm{y} \mid x)P(x)$$
- Expectation $$\mathbb{E}_{\mathrm{x}\sim P}[f(x)] = \sum_x P(x)f(x)$$ or $$\mathbb{E}_{\mathrm{x}\sim p}[f(x)] = \int p(x)f(x)dx$$
- Variance $$\mathrm{Var}(f(x)) = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])^2\right]$$
- Covariance $$\mathrm{Cov}(f(x), g(y)) = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])(g(y) - \mathbb{E}[g(y)])\right]$$
- Covariance matrix $$\begin{array}{cc} & \begin{array}{ccc} \mathrm{x} & \mathrm{y} & \mathrm{z} \end{array} \\ \begin{array}{c} \mathrm{x} \\ \mathrm{y} \\ \mathrm{z} \end{array} & \begin{bmatrix} var(x) & cov(x,y) & cov(x,z) \\ cov(x,y) & var(y) & cov(y,z) \\ cov(x,z) & cov(y,z) & var(z) \end{bmatrix} \end{array}$$

Summary: Probability Distributions

- Uniform distribution $$P(\mathrm{x} = x_i) = \frac{1}{k}$$ or $$u(x; a, b) = \frac{1}{b-a} \quad x \in [a, b]$$
- Bernoulli distribution $$P(\mathrm{x} = 1) = \phi$$ and $$P(\mathrm{x} = 0) = 1 - \phi$$
  - Multinoulli: over multiple states
- Gaussian distribution $$\mathcal{N}(x; \mu, \sigma^2) = \sqrt{\frac{1}{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$ $$\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sqrt{\frac{1}{(2\pi)^n \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$
- Exponential distribution $$p(x; \lambda) = \lambda \mathbf{1}_{x \geq 0} \exp(-\lambda x)$$
- Laplace distribution $$\text{Laplace}(x; \mu, \gamma) = \frac{1}{2\gamma} \exp\left(-\frac{|x-\mu|}{\gamma}\right)$$
- Dirac distribution $$p(x) = \delta(x - \mu)$$



- Empirical data distribution $$\hat{p}(\boldsymbol{x}) = \frac{1}{m} \sum_{i=1}^{m} \delta(\boldsymbol{x} - \boldsymbol{x}^{(i)})$$

Summary: Information Theory

- Self-information *I(x)* of one event: $I(x) = -\log P(x)$

- Shannon Entropy *H(x)* $H(\mathrm{x}) = \mathbb{E}_{\mathrm{x}\sim P}[I(x)] = -\mathbb{E}_{\mathrm{x}\sim P}[\log P(x)]$
  - Average self-information over all outcomes of a random variable
  - Number of bits required to transmit symbols drawn from *P* with a code designed for *P*

- Cross-Entropy *H(P, Q)* $H(P, Q) = -\mathbb{E}_{\mathrm{x}\sim P} \log Q(x)$
  - Number of bits required to transmit symbols drawn from *P* with a code designed for *Q*

- Kullback-Leibler Divergence $D_{KL}(P||Q)$ $D_{\mathrm{KL}}(P\|Q) = \mathbb{E}_{\mathrm{x}\sim P}[\log P(x) - \log Q(x)]$
  - Measures similarity between two distributions *P* and *Q* *= H(P,Q) - H(P)*
  - Note: $D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$

Overview of DL-Lecture